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Class:-SYMCA

Theory:- Menu in Android

A **Menu** in Android is a user interface component that provides options or actions to users. It is typically used to offer navigation, settings, or actions related to the current screen or app context. Menus help enhance app usability by organizing actions in an accessible way.

Types of Menus in Android:

1. **Options Menu** ○ Appears when the user presses the menu button (or more options in the action bar).
 - Commonly used for settings, help, search, etc.
2. **Context Menu** ○ Appears when the user long-presses on a UI element.
 - Offers actions specific to that item (e.g., delete, share, rename).
3. **Popup Menu** ○ A small menu anchored to a view (like a button). ○ Useful for showing related actions in a compact way

ActivityMain:-

```
public class MainActivity extends AppCompatActivity {
```

```
    TextView textView;
```

```
    Button popupButton;
```

```
    @Override    protected void onCreate(Bundle  
savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    setContentView(R.layout.activity_main);    textView  
    = findViewById(R.id.textView);    popupButton =  
    findViewById(R.id.popupButton);
```

```

        // Register context menu for TextView
registerForContextMenu(textView);    //

Popup menu on button click
popupButton.setOnClickListener(v -> {

    PopupMenu popupMenu = new PopupMenu(MainActivity.this, popupButton);
    popupMenu.getMenuInflater().inflate(R.menu.menu_main, popupMenu.getMenu());
    popupMenu.setOnMenuItemClickListener(item -> {

        Toast.makeText(this, "Popup clicked: " + item.getTitle(),
Toast.LENGTH_SHORT).show();

        return true;

    });

    popupMenu.show();

});
}

// Options menu    @Override    public boolean
onCreateOptionsMenu(Menu menu) {
    getMenuInflater().inflate(R.menu.menu_main, menu);

    return true;
}

@Override    public boolean
onOptionsItemSelected(MenuItem item) {

    Toast.makeText(this, "Options clicked: " + item.getTitle(),
Toast.LENGTH_SHORT).show();

    return true;
}

// Context menu

@Override

    public void onCreateContextMenu(ContextMenu menu, View
v, ContextMenu.ContextMenuInfo menuInfo) {
    super.onCreateContextMenu(menu, v, menuInfo);

```

```

menu.setHeaderTitle("Context Menu");    menu.add(0,
v.getId(), 0, "Edit");    menu.add(0, v.getId(), 1, "Delete");
    }

    @Override    public boolean
onContextItemSelected(MenuItem item) {
    Toast.makeText(this, "Context clicked: " + item.getTitle(),
    Toast.LENGTH_SHORT).show();

        return true;
    }
}

```

activity_main.xml:-

```

<LinearLayout

xmlns:android="http://schemas.android.com/apk/res/android"

android:layout_width="match_parent"

android:layout_height="match_parent"

android:orientation="vertical"    android:padding="20dp">

    <TextView        android:id="@+id/textView"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:text="Long Press Me (Context Menu)"

android:textSize="18sp"

android:padding="16dp"

android:background="#E0E0E0" />

    <Button

android:id="@+id/popupButton"

android:layout_width="wrap_content"

```

```
android:layout_height="wrap_content"
android:text="Show Popup Menu"
android:layout_marginTop="30dp"/>
</LinearLayout>
```

menu_main.xml:-

```
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item        android:id="@+id/action_settings"
android:title="Settings"
android:icon="@android:drawable/ic_menu_preferences"
android:showAsAction="ifRoom" />
    <item        android:id="@+id/action_help"
android:title="Help"
android:icon="@android:drawable/ic_menu_help"
android:showAsAction="ifRoom" />
</menu>
```

Output:-

